

Aaron Randolph Arada

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EDUCATION

FEU Institute of Technology

Bachelor of Science in Computer Science with specialization in Software Engineering

Manila, Philippines

Aug. 2024 - Present

PROJECTS

KUMPAS (FSL Translator) | *React, Python, TensorFlow, MediaPipe*

June 2026 | [Live Demo](#)

- Developed a machine-learning system for real-time Filipino Sign Language recognition at ACM TechSprint: Asteria, supporting **29 gesture classes** through webcam input and MediaPipe hand landmarks
- Trained an LSTM sequence classification model with TensorFlow/Keras on **30-frame gesture sequences** containing **126 hand-landmark features per frame**
- Deployed a full-stack translator with a Vercel frontend and Railway Python inference backend, supporting translation output across **6 Philippine languages**

Aaron Intelligence | *React, FastAPI, Supabase, Gemini API*

June 2026 | [Live Demo](#)

- Engineered a RAG-based portfolio assistant indexing **18 Markdown files** into **102 searchable chunks**, enabling natural-language Q&A about projects, skills, education, and experience
- Reduced per-query context by approximately **97%** by retrieving the top **3 of 102 chunks** using Gemini Embedding 001 and Supabase pgvector instead of injecting the full knowledge base
- Integrated a React chat interface with a FastAPI backend and built a **65-question automated evaluation suite** tracking request success, retries, and median/P95 response latency

AlgoVisualizer | *React, JavaScript, CSS*

June 2026

- Integrated an interactive algorithm visualization module into the Aaron Intelligence dashboard, supporting **11 algorithms** across sorting and pathfinding modes
- Implemented step-by-step animations for **7 sorting algorithms** and **4 pathfinding algorithms** on a **13-by-35 grid**, including shortest-path rendering and recursive maze generation
- Added configurable animation speeds, array sizes ranging from **5 to 120 bars**, responsive resizing, and keyboard-accessible navigation between visualization modes

Mirror Mentor (Hackathon Project Lead) | *FastAPI, Supabase, Gemini API, Vercel*

May 2026

- Led backend development for an AI-assisted educational platform during ACM Developers Week, designing and maintaining **17 REST API endpoints** supporting authentication, conversations, and code analysis
- Integrated Google Gemini API with a **4-phase Socratic questioning framework** to deliver guided programming feedback through multi-step code analysis workflows
- Established secure authentication with Supabase Auth and RLS, enabled real-time synchronization through Supabase Realtime, and persisted conversations in PostgreSQL; validated usability with **5 student users**

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++

Frontend: React, Next.js, TailwindCSS, shadcn/ui, HTML/CSS

Backend: FastAPI, Node.js, REST APIs, Supabase, PostgreSQL

AI/ML: TensorFlow/Keras, LSTM, RAG, Gemini API, Sentence Transformers, OpenCV, MediaPipe

Cloud/Tools: AWS EC2, AWS S3, Vercel, Railway, Git, DBeaver, VS Code

CERTIFICATIONS

AWS Academy Generative AI Foundations – AWS Academy, Expected 2026

Pearson IT Specialist: Python – Certiport, Jul. 2026

Microsoft Office Specialist: Excel Associate (Microsoft 365 Apps) – Microsoft, Aug. 2023